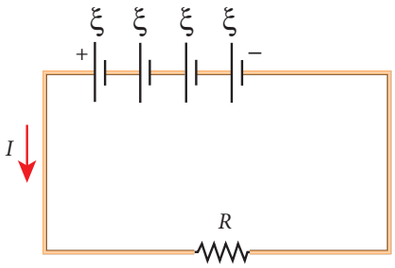


Cella in series (Schematic diagram)



Cells in series (circuit diagram)

Figure 2.21 cells in series

The total emf of the battery = $n\xi$

The total resistance in the circuit = $nr + R$

By Ohm's law, the current in the circuit is

$$I = \frac{\text{total emf}}{\text{total resistance}} = \frac{n\xi}{nr + R} \qquad (2.39)$$

Case (a) If r << R, then,

$$I = \frac{n\xi}{R} \approx nI_1 \qquad (2.40)$$

where, I_1 is the current due to a single cell $\left(I_1 = \frac{\xi}{R}\right)$

Thus, if r is negligible when compared to R the current supplied by the battery is n times that supplied by a single cell.

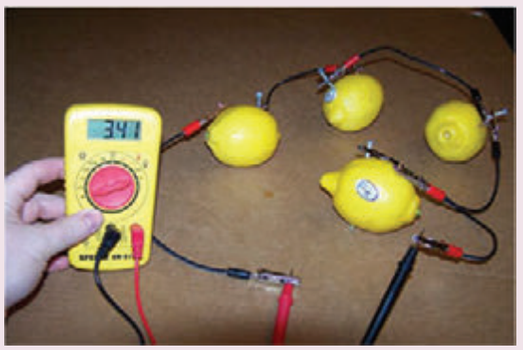
Case (b) If r>>R, $I = \frac{n\xi}{nr} \approx \frac{\xi}{r}$ (2.41)

It is the current due to a single cell. That is, current due to the whole battery is the same as that due to a single cell and hence there is no advantage in connecting several cells.

Thus series connection of cells is advantageous only when the effective internal resistance of the cells is negligibly small compared with R.

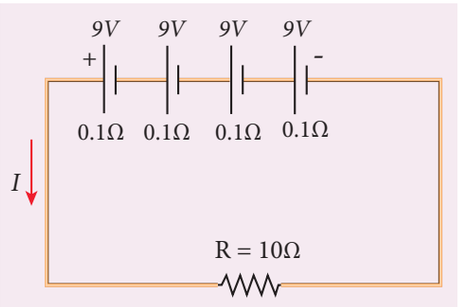
ACTIVITY

Construct lemon cells in series and observe the potential of this combination



EXAMPLE 2.18

From the given circuit,



Find

i) Equivalent emf of the combination

ii) Equivalent internal resistance

iii) Total current

iv) Potential difference across external resistance

v) Potential difference across each cell

Solution

i) Equivalent emf of the combination $\xi_{eq} = n\xi = 4\ 9 = 36$ V

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